UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the June 2005 question paper

5054 PHYSICS

5054/04 **Paper 4 (Alternative to Practical), maximum mark 30**

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UNIVERSITY of CAMBRIDGE
International Examinations

June 2005

GCE O Level

MARK SCHEME

MAXIMUM MARK: 30

SYLLABUS/COMPONENT: 5054/04

PHYSICS
(Alternative to Practical)

UNIVERSITY of CAMBRIDGE
International Examinations

1 **(a)** values calculated correctly mass (to 1 dp); volume (max 1 if units in table) **[2]**

(b) axes, correct way round, labelled quantity and unit
scales; more than ½ page, sensible, include (0,0)
6 points plotted ± 1 square (ignore 0,0)
best fit straight line drawn, neatly (through minimum 6 points) **[4]**

(c) triangle drawn/values more than ½ line length/½ points from table
values accurately computed (allow any relevant values)
minimum 2 sf and correct unit **[3]**

(d) correct glass type identified for their value **[1]**

(e) **(i)** water would increase mass (cause problem)/time taken to dry marbles **[1]**

(ii) large enough to contain marbles/will not overflow/enough to cover marbles/
suitable values quoted e.g. 40 cm^3 water or 53.5 cm^3 **[1]**

(f) micrometer/vernier calliper/ruler only if >one marble in a line
diameter of the marble
conversion r to d and substitution/equation changed to d not r
(can back-credit diameter here if blank or radius is given above) **[3]**

Total: 15

2 **(a)** circuit drawn, A in series with lamp and rheostat
V in parallel with lamp **[2]**

(b) table with three columns, heading current, voltage, resistance
ignore repeats three correct units **[2]**

(c) No: filament still has resistance (when no current flows) **[1]**

Total: 5

3 **(a)** to give a sufficient temperature rise/heat up the lead **[1]**

(b) to avoid breaking the thermometer **[1]**

(c) advantage fewer inversions needed (for same height)/larger $\Delta\theta$ for the same number of inversions
more accurate/thermal energy/potential energy

disadvantage difficult to invert quickly/lead shot more likely to slide/longer time taken/tube or bung may be damaged/more heat loss **[2]**

(d) **(i)** 345 (no unit required, ignore incorrect unit) **[1]**

(ii) height fallen by shot smaller than measured length of tube/some energy lost to tube or bung/error in specified reading **[1]**

Total: 6

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4 **(a)** any two from:

number/weight of paperclips
length of stem height dropped
stem to wings ratio surface area of wing
paper weight **[2]**

(b) longer wings, increases time (comparison needed) **[1]**

(c) sensible suggestion, e.g. use marker to fix drop height/repeats and average hold/drop in the same way/use stopwatch **[1]**

Total: 4

Paper total 30 marks

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